

Harrow Bay – Geotechnical Survey Summary

Integrated ground model and design parameters

Sofie Lindqvist · 14 March 2025 · HB-GEO-RPT-0011

Summary

This document summarises the results of the Harrow Bay geotechnical site investigation (offshore campaign May–August 2024, laboratory testing until December 2024) for the 40 wind turbine positions, the offshore substation HB-OSS and the export cable route. It is the geotechnical input to HB-ENG-SPC-0042 Monopile Design Basis and to the pile drivability assessment.

Scope	Quantity
Seabed CPTs at turbine positions	40 (one per position) + 9 contingency
Geotechnical boreholes (to 60 m below seabed)	8 (one per soil province plus OSS)
Cable route vibrocores	46 along 62 km export route
Laboratory tests (classification, triaxial, cyclic DSS, thermal)	1,214



Geotechnical drilling vessel on site, July 2024

1. Ground model

The site is underlain by a sequence typical of the southern North Sea: a veneer of mobile Holocene sand (0–3 m), over dense to very dense Pleistocene sand of the Bolders Bank / Botney Cut formations, over stiff to very stiff glacial clay. Four soil provinces were defined from CPT and geophysical data:

Province	Positions	Typical profile	Comment
A – Sand dominated	WTG 01–14	Dense sand to 25 m, stiff	High driving resistance

		clay below	
B – Layered	WTG 15–27	Interbedded sand and clay, 2–6 m layers	Most variable province
C – Clay dominated	WTG 28–36, OSS	Stiff clay from 4 m, cu 150–400 kPa	Lower stiffness, longer piles
D – Channel infill	WTG 37–40	Soft to firm clay in buried channel to 18 m	Design driver for embedment

Table 1 – Soil provinces



Core samples from borehole BH-05 (province C) in the laboratory

2. Key parameters for monopile design

Parameter	Province A	Province B	Province C	Province D
Design embedment (m)	31–33	33–35	35–37	38–40
Peak qc in sand (MPa)	45–70	30–55	–	–
Undrained shear strength (kPa)	–	100–250	150–400	35–90 (infill)
Small-strain stiffness Gmax (MPa)	180–260	120–220	90–160	40–90

With a monopile diameter of 9.5 m and water depths of 28–36 m, design embedment lengths range from 31 m (province A) to 40 m (province D). Total pile lengths range from 82 m to 96 m; heaviest pile 1,640 t.

3. Installation risks

- Pile refusal risk in province A: dense sand with $q_c > 60$ MPa at WTG 03, 07 and 11. Driveability study shows blow counts up to 260 blows/0.25 m with the IHC S-4000 hammer – acceptable, but with little margin.
- Boulders: low to moderate risk, see geophysical summary; two positions micro-sited.

- Pile run risk in province D: soft channel infill – controlled with hammer energy limits for the first 8 m.
- Noise: dense sand increases hammer energy and noise; bubble curtains are mandatory for all piling (consent condition).

4. Export cable route

Seabed along the 62 km route is predominantly sand and gravel. Mobile sand waves up to 4 m high occur between KP 18 and KP 27; pre-sweeping is planned. Thermal resistivity of the seabed sediments is 0.6–0.9 K·m/W (favourable for cable rating).

5. Open items

- Confirm cyclic degradation parameters for province D with the certification body (target Q4 2026).
- Update ground model with as-driven pile data after the 2026 campaign (Oliver Grant).